

E-COMMERCE SALES ANALYSIS USING SQL

A Data Analysis Project

Tools & Technologies:

SQL | MySQL | Data Analysis

Dataset:

E-Commerce Sales Dataset

Executive Summary

This project analyzes e-commerce sales data using SQL to understand overall business performance across products, categories, regions, sales, quantity, and profit.

The dataset contains **3,500 sales records** covering the period from **2022 to 2024**. The analysis focuses on identifying top-performing products, evaluating category and regional sales, measuring profitability, and understanding sales trends over time.

The project demonstrates how SQL can be used to transform raw sales transaction data into meaningful business insights that can support product planning, regional strategy, revenue optimization, and data-driven decision-making.

OBJECTIVE

The main objective of this project is to analyze e-commerce sales data using SQL and convert raw transactional data into meaningful business insights.

The analysis focuses on:

- Identifying the top-selling products based on quantity sold.
- Analyzing total sales across different categories.
- Comparing sales performance across regions.
- Calculating total profit generated by products and categories.
- Identifying the most profitable products.
- Analyzing monthly and yearly sales trends.

- Comparing quantity sold across products.
- Identifying the highest-revenue categories.
- Evaluating regional sales and profit performance.
- Calculating average sales and profit.
- Ranking products based on sales and profit.
- Finding the best-performing products within each category.
- Analyzing the relationship between sales and profit.
- Using SQL aggregation and window functions for business analysis.
- Creating queries that can support an e-commerce sales dashboard.

The overall goal is to demonstrate how SQL can be applied to real-world sales data for business reporting and data-driven decision-making.

ABOUT THE DATASET

This project uses an e-commerce sales dataset containing information about order dates, products, categories, regions, quantity sold, sales revenue, and profit.

The dataset contains:

- **3,500 records**
- **10 unique products**
- **3 categories**
- **4 regions**
- Order dates ranging from **2022 to 2024**

Important Columns

- **Order_Date** – Date on which the order was placed.
- **Productname** – Name of the product sold.
- **Category** – Category to which the product belongs.
- **Region** – Geographical region of the sale.

- **Quantity** – Number of units sold.
- **Sales** – Revenue generated from the sale.
- **Profit** – Profit generated from the sale.

DATASET OVERVIEW

Overall Business Performance

- **Total Quantity Sold:** 17,261 units
- **Total Sales:** 10,667,881
- **Total Profit:** 1,844,665.21
- **Data Period:** 2022–2024

Category Performance

The dataset contains three major categories:

- Electronics
- Accessories
- Office

Electronics is the largest category by sales, followed by **Accessories** and **Office**.

Regional Performance

The four regions included in the dataset are:

- North
- South
- East
- West

Among these regions, **West** generates the highest total sales.

ANALYTIC QUESTIONS

1. Find the total sales for each region.

```
SELECT
    region, SUM(sales) AS total_sales
FROM
    ecommerce.ecommerce_sales_data
GROUP BY region
ORDER BY total_sales DESC;
```

	region	total_sales
▶	West	2844450
	East	2675110
	South	2659548
	North	2488773

2. Find the total profit for each region.

```
SELECT
    region, SUM(profit) AS total_profit
FROM
    ecommerce.ecommerce_sales_data
GROUP BY region
ORDER BY total_profit DESC;
```

	region	total_profit
▶	West	495358.72999999917
	East	464888.46
	South	458103.27000000001
	North	426314.74999999924

3. Find total sales and profit for each category.

```
SELECT
    category,
    SUM(profit) AS total_profit,
    SUM(sales) AS total_sales
FROM
    ecommerce.ecommerce_sales_data
GROUP BY category
ORDER BY total_sales , total_profit DESC;
```

	category	total_profit	total_sales
▶	Office	185394.88	1094216
	Accessories	736084.7400000017	4247591
	Electronics	923185.5900000011	5326074

4. Find the total quantity sold for each region.

```
SELECT
    region, SUM(quantity) AS total_quantity
FROM
    ecommerce.ecommerce_sales_data
GROUP BY region
ORDER BY total_quantity DESC;
```

	region	total_quantity
▶	West	4488
	South	4361
	East	4306
	North	4106

5. Find the number of order in each region.

```
SELECT
    region, COUNT(*) AS number_of_order
FROM
    ecommerce.ecommerce_sales_data
GROUP BY region
ORDER BY number_of_order DESC;
```

	region	number_of_order
▶	West	898
	South	883
	East	861
	North	858

6. Find the average sales for each category.

```
SELECT
    category, AVG(sales) AS avg_sales
FROM
    ecommerce.ecommerce_sales_data
GROUP BY category
ORDER BY avg_sales DESC;
```

	category	avg_sales
►	Office	3065.0308
	Electronics	3057.4478
	Accessories	3031.8280

7. Find the average profit for each region.

```
SELECT
    region, ROUND(AVG(profit), 2) AS avg_profit
FROM
    ecommerce.ecommerce_sales_data
GROUP BY region
ORDER BY avg_profit DESC;
```

	region	avg_profit
►	West	551.62
	East	539.94
	South	518.8
	North	496.87

8. Find region where total sales are greater than the average region sales.

```
SELECT
    region, SUM(sales) AS total_sales
FROM
    ecommerce.ecommerce_sales_data
GROUP BY region
HAVING SUM(sales) > (SELECT
    AVG(regional_sales)
FROM
    (SELECT
        SUM(sales) AS regional_sales
    FROM
        ecommerce.ecommerce_sales_data
    GROUP BY region) AS region_totals);
```

	region	total_sales
►	East	2675110
	West	2844450

9. Find category where total profit is greater than 1,00,000.

```
SELECT
    category, SUM(profit) AS total_profit
FROM
    ecommerce.ecommerce_sales_data
WHERE
    category < 100000
GROUP BY category;
```

	category	total_profit
▶	Office	185394.88
	Accessories	736084.7400000017
	Electronics	923185.5900000011

10. Find top 10 product base on total sales.

```
SELECT
    productname, SUM(sales) AS total_sales
FROM
    ecommerce.ecommerce_sales_data
GROUP BY productname
ORDER BY total_sales DESC
LIMIT 10;
```

	productname	total_sales
▶	Camera	1177381
	Monitor	1160048
	Printer	1094216
	Mouse	1074398
	Smartphone	1069681
	Smartwatch	1049211
	Keyboard	1024507
	Tablet	1023928
	Laptop	1005873
	Headphones	988638

11. Find the top 5 product based on total profit.

```
SELECT
    productname, SUM(profit) AS total_profit
FROM
    ecommerce.ecommerce_sales_data
GROUP BY productname
ORDER BY total_profit DESC
LIMIT 5;
```

	productname	total_profit
▶	Camera	207630.98999999999
	Monitor	202028.16999999993
	Mouse	185763.68999999992
	Laptop	185756.81000000014
	Printer	185394.88

12. Find the region with the highest total sales.

```
SELECT
    region, SUM(sales) AS total_sales
FROM
    ecommerce.ecommerce_sales_data
GROUP BY region
ORDER BY total_sales DESC
LIMIT 1;
```

	region	total_sales
▶	West	2844450

13. Find the region with the highest total profit.

```
SELECT
    region, SUM(profit) AS total_profit
FROM
    ecommerce.ecommerce_sales_data
GROUP BY region
ORDER BY total_profit DESC
LIMIT 1;
```

	region	total_profit
►	West	495358.72999999917

14. Find the product with the highest quantity sold.

```
SELECT
    productname, SUM(quantity) AS quantity_sold
FROM
    ecommerce.ecommerce_sales_data
GROUP BY Productname
ORDER BY quantity_sold DESC
LIMIT 1;
```

	productname	quantity_sold
►	Monitor	1876

15. Find total sales by year.

```
SELECT
    YEAR(STR_TO_DATE(Order_Date, '%d-%m-%Y')) AS year,
    SUM(Sales) AS total_sales
FROM
    ecommerce.ecommerce_sales_data
GROUP BY YEAR(STR_TO_DATE(Order_Date, '%d-%m-%Y'))
ORDER BY year DESC;
```

	year	total_sales
►	2024	3625319
	2023	3786592
	2022	3255970

16. Find total profit by year.

```
SELECT
    YEAR(STR_TO_DATE(order_date, '%d-%m-%y')) AS year,
    SUM(profit) AS total_profit
FROM
    ecommerce.ecommerce_sales_data
GROUP BY year
ORDER BY year DESC;
```

	year	total_profit
▶	2020	1844665.2099999993

17. Find the year with highest sales.

```
SELECT
    YEAR(STR_TO_DATE(order_date, '%d-%m-%y')) AS year,
    SUM(sales) AS total_sale
FROM
    ecommerce.ecommerce_sales_data
GROUP BY year
ORDER BY year DESC
LIMIT 1;
```

	year	total_sale
▶	2020	10667881

18. Find the year with highest profit.

```
SELECT
    YEAR(STR_TO_DATE(order_date, '%d-%m-%y')) AS year,
    SUM(profit) AS total_profit
FROM
    ecommerce.ecommerce_sales_data
GROUP BY year
ORDER BY year DESC
LIMIT 1;
```

	year	total_profit
▶	2020	1844665.2099999993

19. Find month sales performance.

```
SELECT
    MONTH(STR_TO_DATE(order_date, '%d-%m-%y')) AS month,
    SUM(sales) AS total_sales
FROM
    ecommerce.ecommerce_sales_data
GROUP BY month
ORDER BY month;
```

	month	total_sales
▶	1	967614
	2	702414
	3	939614
	4	821876
	5	1034687
	6	839805
	7	844940
	8	924939
	9	885546
	10	892269
	11	855075
	12	959102

20. Calculate the profit percentage for each order using.

```
SELECT
    productname,
    profit,
    sales,
    (profit / sales) * 100 AS profit_percentage
FROM
    ecommerce.ecommerce_sales_data;
```

	productname	profit	sales	profit_percentage
	Laptop	1219.73	7544	16.1682131495228
	Keyboard	1871.88	6360	29.432075471698116
	Smartphone	485.75	2428	20.006177924217464
	Headphones	38.52	574	6.710801393728223
	Keyboard	660.47	6744	9.793446026097271
	Headphones	213.4	798	26.741854636591476
	Printer	339	2860	11.853146853146853
	Printer	1129.77	5410	20.882994454713494
	Laptop	580.15	4445	13.051743532058493
	Smartwatch	1197.09	7536	15.884952229299362
	Monitor	263.26	4165	6.320768307322928 ...

21. Rank all product based on total sales using rank().

```
select productname, sum(sales) as total_sales,  
rank()  
over(order by sum(sales) desc)  
as ranked_sales  
from ecommerce.ecommerce_sales_data  
group by productname;
```

	productname	total_sales	ranked_sales
▶	Camera	1177381	1
	Monitor	1160048	2
	Printer	1094216	3
	Mouse	1074398	4
	Smartphone	1069681	5
	Smartwatch	1049211	6
	Keyboard	1024507	7
	Tablet	1023928	8
	Laptop	1005873	9
	Headphones	988638	10

22. Rank all region based on total profit using dense rank().

```
select region, sum(profit) as total_profit,  
dense_rank()  
over(order by sum(profit) desc)  
as ranked_profit  
from ecommerce.ecommerce_sales_data  
group by region;
```

	region	total_profit	ranked_profit
▶	West	495358.729999999917	1
	East	464888.46	2
	South	458103.27000000001	3
	North	426314.749999999924	4

23. Find the top 3 product with the cheapest sale in each category.

```
SELECT *  
from(select productname,category,sales,  
rank()  
over(partition by category  
      order by sales desc)  
as sales_rank  
from ecommerce.ecommerce_sales_data  
) as ranked  
where sales_rank <= 3;
```

	productname	category	sales	sales_rank
▶	Keyboard	Accessories	10782	1
	Monitor	Accessories	10665	2
	Keyboard	Accessories	10629	3
	Laptop	Electronics	10773	1
	Laptop	Electronics	10728	2
	Laptop	Electronics	10620	3
	Printer	Office	10656	1
	Printer	Office	9528	2
	Printer	Office	9477	3

24. Find the highest selling product in each region.

```
select *  
from(  
select productname,region,sum(sales)as total_sales,  
rank()  
over (partition by region  
      order by sum(sales) asc)  
as selling_product  
from ecommerce.ecommerce_sales_data  
group by Productname,Region  
)as ranked  
where selling_product <= 1;
```

	productname	region	total_sales	selling_product
▶	Mouse	East	234142	1
	Headphones	North	198684	1
	Laptop	South	207327	1
	Keyboard	West	235583	1

25. Find the most profitable product in each category.

```
select *
from (
select productname,category,sum(profit) as total_profit,
rank()
over (partition by category
      order by sum(profit) desc)
as profitable_product
from ecommerce.ecommerce_sales_data
group by productname,category
)as racked
where profitable_product <=1;
```

	productname	category	total_profit	profitable_product
▶	Monitor	Accessories	202028.169999999993	1
	Camera	Electronics	207630.989999999999	1
	Printer	Office	185394.88	1

26. Find product whose total sales are greater than the average product sale.

```
SELECT
    productname, SUM(sales) AS total_sales
FROM
    ecommerce.ecommerce_sales_data
GROUP BY productname
HAVING SUM(sales) > (SELECT
    AVG(total_sales)
    FROM
        (SELECT
            SUM(sales) AS total_sales
        FROM
            ecommerce.ecommerce_sales_data
        GROUP BY productname) AS product_sales);
```

	productname	total_sales
▶	Printer	1094216
	Mouse	1074398
	Camera	1177381
	Monitor	1160048
	Smartphone	1069681

27. Find the cumulative sales overtime using window function.

```
select order_date,sales,
sum(sales) OVER(
order by str_to_date(order_date,'%d-%m-%y')
)as cumulative_sales
from ecommerce.ecommerce_sales_data
order by str_to_date(order_date,'%d-%m-%y');
```

	order_date	sales	cumulative_sales
▶	01-01-2022	3474	43400
	01-01-2024	5790	43400
	01-01-2023	1405	43400
	01-01-2023	5904	43400
	01-01-2022	1296	43400
	01-01-2023	4470	43400
	01-01-2022	6372	43400
	01-01-2022	9234	43400
	01-01-2023	1514	43400
	01-01-2022	3941	43400
	02-01-2024	3624	81722
	02-01-2022	1240	81722

...

28. Calculate running total of profit of region.

```
select order_date,profit,region,
sum(profit) over(
partition by region
order by str_to_date(order_date,'%d-%m-%y')
)running_profit
from ecommerce.ecommerce_sales_data
order by region,str_to_date(order_date,'%d-%m-%y');
```

	order_date	profit	region	running_profit
▶	01-01-2022	570.75	East	1135.75
	01-01-2022	565	East	1135.75
	03-01-2023	701.11	East	1913.9900000000002
	03-01-2023	77.13	East	1913.9900000000002
	04-01-2024	42.37	East	3444.76
	04-01-2022	1488.4	East	3444.76
	05-01-2022	720.08	East	6536.72
	05-01-2023	1923.89	East	6536.72
	05-01-2024	447.99	East	6536.72
	06-01-2023	57.51	East	7675.76
	06-01-2023	1081.53	East	7675.76

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